

### 2.1.2 Compounds, formulae and equations

| Learning outcomes                                                                                                                                                                                                                                                                                                                                                                            | Additional guidance                                                                                                                                                                  |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |
| <b>Formulae and equations</b>                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |
| <b>(a)</b> the writing of formulae of ionic compounds from ionic charges, including:<br><b>(i)</b> prediction of ionic charge from the position of an element in the periodic table<br><b>(ii)</b> recall of the names and formulae for the following ions: $\text{NO}_3^-$ , $\text{CO}_3^{2-}$ , $\text{SO}_4^{2-}$ , $\text{OH}^-$ , $\text{NH}_4^+$ , $\text{Zn}^{2+}$ and $\text{Ag}^+$ | Note that 'nitrate' and 'sulfate' should be assumed to be $\text{NO}_3^-$ and $\text{SO}_4^{2-}$ .<br><br>Charges on ions other than in <b>(i)</b> and <b>(ii)</b> will be provided. |
| <b>(b)</b> construction of balanced chemical equations (including ionic equations), including state symbols, for reactions studied and for unfamiliar reactions given appropriate information.                                                                                                                                                                                               | <i>MO.2</i>                                                                                                                                                                          |